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A Note on

A toy model of unemployment

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1. Economic question and literature

Why does employment adjust gradually after a shock, and why can unemployment move much more than productivity? A frictionless labor market allocates hours through a wage that equates supply and demand. With costly recruitment, a firm instead compares the cost of a new hire with the current and future value of an employment relationship. The wage divides that surplus between the worker and the firm. Productivity, bargaining power and the cost of financing capital can therefore affect hiring through different channels.

Mortensen and Pissarides [1994] develop the job-creation and job-destruction approach to unemployment. Merz [1995] embeds search in a real-business-cycle framework, showing that the connection between labor-market frictions and aggregate fluctuations predates the 2000s. Shimer [2005] documents the difficulty of a standard calibrated search model in generating observed unemployment and vacancy volatility. Hall [2005] highlights wage determination as a key part of that propagation problem.

The present exercise uses a compact hiring-cost mechanism rather than a full vacancy-matching system. Its closest modeling motivation is the tractable unemployment framework of Blanchard and Galí [2010]. The firm pays a convex cost to expand hiring, employment relationships survive with a fixed probability and wages satisfy a Nash sharing rule. This retains a distinction between the marginal product of a worker and the wage, which matters for both hiring and the response of unemployment.

Gertler and Trigari [2009] introduce staggered Nash wage bargaining in an aggregate model. Christiano et al. [2016] study unemployment and business cycles in a quantitatively richer framework. These contributions motivate comparing mechanisms and data rather than assuming that an estimated shock process uniquely identifies a labor-market friction. Here the student must decide which observables can discipline recruitment, separations and bargaining, and must distinguish a successful numerical solution from empirical adequacy.

2. The model

Time is quarterly. The labor force has unit mass, n_t is employment and $u_t = 1-n_t$ is current unemployment. The separation rate is ρ. The recruitment variable x_t is hires per current unemployed worker, not a transition probability. This distinction preserves the supplied timing and calibration. Individual firms and families take the aggregate pool u_t as given, and aggregation imposes $u_t = 1 - n_t$.

2.1 Households

A representative family pools its members' consumption risk and maximizes

$$\max_{\{c_{j,t+\tau},b_{j,t+\tau},n_{j,t+\tau},i_{j,t+\tau},k_{j,t+\tau}\}_{\tau\geq 0}} E_t \sum_{\tau\geq 0} \beta^\tau \left[\varepsilon^C_{t+\tau} \frac{c_{j,t+\tau}^{1-\sigma_C}}{1-\sigma_C} - \chi n_{j,t+\tau}\right]. \tag{1}$$

Use the logarithmic limit at $\sigma_C = 1$. Each family has unit mass, so

$$n_{jt} = 1 - u_{jt}. \tag{2}$$

The employment accounting consistent with the firms below is

$$n_{jt} = (1-\rho)n_{j,t-1} + x_t u_t. \tag{3}$$

At the aggregate level, new hires are $x_t u_t$. Using lagged unemployment instead would define a different timing convention and require a different model and calibration.

The real budget is

$$c_{jt} + i_{jt} + f(i_{jt}/i_{j,t-1})i_{j,t-1} + b_{jt} = r_{t-1}b_{j,t-1} + n_{jt}w_t + z_t k_{j,t-1} + (1-n_{jt})\bar{b} + \Pi_{jt} - T_{jt}. \tag{4}$$

Here r_t is a gross real bond return, z_t the capital rental rate and $\bar{b}$ the fixed unemployment benefit. Firms' profits Π_{jt} are rebated to their owners and treated as lump-sum income. They are required for aggregate accounting when hiring costs create a gap between output and factor payments. Investment adjustment costs are $f(v) = \kappa(v-1)^2/2$. Capital obeys

$$\varepsilon^I_t i_{jt} = k_{jt} - (1-\delta)k_{j,t-1}. \tag{5}$$

Dropping the family index, let $\mathcal{B}_t$ be income minus expenditure, $\mathcal{N}_t = (1-\rho)n_{t-1} + x_t u_t - n_t$, and $\mathcal{K}_t = \varepsilon^I_t i_t + (1-\delta)k_{t-1} - k_t$. The Lagrangian is

$$\mathcal{L}_t = E_t \sum_{\tau\geq 0} \beta^\tau \left[\varepsilon^C_{t+\tau} \frac{c_{t+\tau}^{1-\sigma_C}}{1-\sigma_C} - \chi n_{t+\tau} + \lambda^C_{t+\tau}\mathcal{B}_{t+\tau} + \lambda^C_{t+\tau}V^H_{t+\tau}\mathcal{N}_{t+\tau} + \lambda^C_{t+\tau}q_{t+\tau}\mathcal{K}_{t+\tau}\right].$$

The multipliers V^H_t and q_t are expressed in consumption units. With $v_t = i_t/i_{t-1}$, the first-order conditions are

$$\lambda^C_t = \varepsilon^C_t c_t^{-\sigma_C}, \tag{6}$$

$$\lambda^C_t = \beta r_t E_t \lambda^C_{t+1}, \tag{7}$$

$$V^H_t = w_t - \bar{b} - \chi/\lambda^C_t + \beta(1-\rho)E_t[(\lambda^C_{t+1}/\lambda^C_t)V^H_{t+1}], \tag{8}$$

$$q_t\varepsilon^I_t = 1 + f'(v_t) + \beta E_t\left[\frac{\lambda^C_{t+1}}{\lambda^C_t}\{f(v_{t+1}) - v_{t+1}f'(v_{t+1})\}\right], \tag{9}$$

$$q_t = \beta E_t[(\lambda^C_{t+1}/\lambda^C_t)(z_{t+1} + (1-\delta)q_{t+1})]. \tag{10}$$

The future investment term follows from differentiating $f(i_{t+1}/i_t)i_t$. It includes both the scale and the ratio effects. At first order around $v = 1$, the investment condition reduces to the equation in the starter. The worker surplus compares the wage with the benefit and the consumption-equivalent disutility of employment.

2.2 Firms

Production uses inherited capital and current workers:

$$y_{it} = \varepsilon_t^A k_{i,t-1}^{\alpha} n_{it}^{1-\alpha}. \tag{11}$$

Employment and profits obey

$$n_{it} = (1-\rho) n_{i,t-1} + x_{it} u_t, \tag{12}$$

$$\Pi_{it} = y_{it} - w_t n_{it} - z_t k_{i,t-1} - G(x_{it}) u_t, \tag{13}$$

where

$$G(x) = \frac{\Gamma}{1+\phi} x^{1+\phi}.$$

Thus ϕ is the elasticity of the marginal hiring cost $G'(x)$, whereas the elasticity of total cost $G(x)$ is $1+\phi$. Recruiting expenditure may include screening, administration and training. It is a real resource cost.

The firm maximizes discounted profits subject to employment accumulation:

$$\mathcal{L}_t^F = E_t \sum_{\tau \geq 0} \beta^{\tau} \frac{\lambda_{t+\tau}^C}{\lambda_t^C} \left[\Pi_{i,t+\tau} + V_{t+\tau}^P \{ (1-\rho) n_{i,t-1+\tau} + x_{i,t+\tau} u_{t+\tau} - n_{i,t+\tau} \} \right].$$

The firm uses the household discount factor because households own it. Its conditions are

$$\begin{aligned} V_t^P &= (1-\alpha) y_{it}/n_{it} - w_t + \beta(1-\rho) E_t[(\lambda_{t+1}^C/\lambda_t^C) V_{t+1}^P], \\ G'(x_{it}) &= V_t^P, \\ z_t &= \alpha y_{it}/k_{i,t-1}. \end{aligned}$$

Combining the first two equations gives the hiring no-arbitrage condition

$$G'(x_{it}) = (1-\alpha) y_{it}/n_{it} - w_t + \beta(1-\rho) E_t[(\lambda_{t+1}^C/\lambda_t^C) G'(x_{i,t+1})].$$

A new relationship is valuable because it produces a current operating surplus and may survive into future periods. Hiring is optimal when its marginal cost equals that value.

2.3 Wage bargaining

Period-by-period Nash bargaining maximizes the joint surplus

$$S_t = (V_t^H)^{\eta} (V_t^P)^{1-\eta}, \qquad \max_{w_t} S_t,$$

where $\eta \in (0,1)$ is the worker's bargaining weight. Taking continuation values as given in the current negotiation, $\partial V_t^H / \partial w_t = 1$ and $\partial V_t^P / \partial w_t = -1$, hence

$$\frac{\partial S_t}{\partial w_t} = 0 \quad \Longleftrightarrow \quad \eta V_t^P = (1-\eta) V_t^H.$$

With a bargaining disturbance the sharing rule becomes

$$\varepsilon_t^M \eta V_t^P = (1 - \varepsilon_t^M \eta) V_t^H.$$

This is a local specification around $\varepsilon^M = 1$, with admissible weights between zero and one. For $\varepsilon_t^M = 1$, substitution yields

$$w_t = \eta \left[(1-\alpha) y_t/n_t + E_t(m_{t,t+1} V_{t+1}^P) \right] + (1-\eta) \left[\bar{b} + \chi/\lambda_t^C - E_t(m_{t,t+1} V_{t+1}^H) \right],$$

where $m_{t,t+1} = \beta(\lambda_{t+1}^C/\lambda_t^C)(1-\rho)$. Benefits and the disutility of work raise the worker's reservation component. The continuation value of employment lowers the current wage needed to attain a given worker surplus. Both signs matter for the interpretation.

2.4 Authorities

Government purchases are $G_t = g^y \bar{Y} \varepsilon_t^G$. Public bonds and taxes satisfy

$$r_{t-1} b_{t-1} + G_t + \bar{b} u_t = b_t + T_t.$$

Benefits are transfers rather than resource costs. Their financing is lump sum in this baseline. The symbol $\bar{b}$ denotes the benefit, not public debt b_t.

2.5 Equilibrium conditions

After aggregating budgets and rebated profits,

$$Y_t = C_t + I_t + f(I_t/I_{t-1}) I_{t-1} + G_t + G(x_t) u_t, \tag{14}$$

$$u_t = 1 - n_t. \tag{15}$$

The investment adjustment cost has zero first derivative at the steady state, so it drops out of the first-order resource constraint. Hiring costs do not: their linear contribution must be divided by $\bar{Y}$ when output is written as a log deviation. The calibration sets $G(\bar{x})\bar{u} = .01\bar{Y}$.

3. Simulation and empirical starting point

The file `unemployment_RBC.mod` is log-linear. Its parameter `delta_N` is the separation rate ρ. It uses $\rho = .1$, $\eta = .5$, $\phi = .15$, $\beta = .993$, $\delta = .025$, $\alpha = .30$, $g^y = .2$, $\sigma_C = 1$, $\kappa = 4$, $\bar{u} = .075$, and $\bar{b} = .5\bar{w}$. The scale Γ matches hiring costs to one percent of output and χ supports the targeted unemployment rate. The implied $\bar{x} = \rho(1 - \bar{u})/\bar{u}$ exceeds one, confirming that it cannot be interpreted as a job-finding probability under this timing. All five shock persistence parameters are .95. For $s \in \{A, G, C, M, I\}$, $\log \varepsilon_t^s = \rho_s \log \varepsilon_{t-1}^s + \eta_t^s$. The code uses their first-order deviations.

The Nash equation retains both worker and firm surpluses. The supplied unit innovation standard deviations normalize first-order responses and should not be interpreted as admissible large changes in a bargaining probability. Simulation checks do not identify an estimation specification.

The downloader supplies raw US output and unemployment responses. Unemployment is a rate, while output is a level or volume series. Define units, seasonal adjustment, sample dates and transformations before selecting measurement equations. Real wages or investment require additional documented series. The assessment leaves data construction, priors and inference to the student, without providing corrected observations or posterior answers.

References

Olivier Blanchard and Jordi Galí. Labor Markets and Monetary Policy: A New Keynesian Model with Unemployment. *American Economic Journal: Macroeconomics*, 2(2):1–30, 2010. doi: 10.1257/mac.2.2.1.

Lawrence J. Christiano, Martin S. Eichenbaum, and Mathias Trabandt. Unemployment and Business Cycles. *Econometrica*, 84(4):1523–1569, 2016. doi: 10.3982/ecta11776.

Mark Gertler and Antonella Trigari. Unemployment Fluctuations with Staggered Nash Wage Bargaining. *Journal of Political Economy*, 117(1):38–86, 2009. doi: 10.1086/597302.

Robert E Hall. Employment Fluctuations with Equilibrium Wage Stickiness. *American Economic Review*, 95(1): 50–65, 2005. doi: 10.1257/0002828053828482.

Monika Merz. Search in the labor market and the real business cycle. *Journal of Monetary Economics*, 36(2): 269–300, 1995. doi: 10.1016/0304-3932(95)01216-8.

D. T. Mortensen and C. A. Pissarides. Job Creation and Job Destruction in the Theory of Unemployment. *The Review of Economic Studies*, 61(3):397–415, 1994. doi: 10.2307/2297896.

Robert Shimer. The Cyclical Behavior of Equilibrium Unemployment and Vacancies. *American Economic Review*, 95(1):25–49, 2005. doi: 10.1257/0002828053828572.